

台州通禾流体控制股份有限公司

Taizhou Tonhe Flow Control Co., Ltd

Tonhe A20 Series Proportional actuator ball valve



➤ Technical Parameters

Valve size 阀体口径	□NPT/BSP 1/4" 3/8" 1/2" 3/4" 1" (Optional)
Maximum working pressure 最大工作压力	1.0MPa
Circulation medium 介质	Fluid, air,water 液体, 空气, 水
Rated voltage 额定电压	DC5V
Protocol 通信协议	LORAWAN 868 MHZ 915 MHZ
Wiring diagram 接线图	2 wire controll 2 线控制
Open/close time 开关时间	≤5S/90°
Life time 工作寿命	70000 times
Valve material 阀体材质	□Brass □SS304 (Optional) □黄铜 □不锈钢304 可选
Actuator material 执行器材料	PPO Engineering Plastics
Sealing material 密封材质	FKM & PTFE
Actuator rotation 执行器旋转角度	90°
Max. torque force 最大扭力	3 N.M
Cable Length 电线长度	0.5m
Environment temperature 环境温度	-15℃~60℃
Liquid temperature 液体温度	2℃~90℃
Manual override 手动功能	No
Indicator 开关指示	Yes
Protection class 防护等级	IP67
Product certificate 认证	ROHS
Company system certification 公司体系认证	ISO 9001:2015
Locked-rotor Current 堵转电流	500mA

The following code are shared 共享编码

deveui	
JoinEUI	70B3D57ED0036D57
AppKey	B3CAD3EF63A34E256918F15862EFE150

Instructions 使用说明

阀门接收 Valve receive	端口 PORT	数据 DATA	备注 Note
	0x0A	0X01	设置开阀 set valve to open
		0X02	设置关阀 set valve to close
		0x00	无效数据，用于获取服务器下发数据 Invalid data, used to obtain data issued by the server
阀门发送 Valve send	0X0A	0XC1	已经开阀 valve has been opened
		0XC2	正在开阀 valve is being opened
		0XC3	已经关阀 valve has been closed
		0XC4	正在关阀 valve is being closed
		0XC5	堵转重试/3 次重试 blocked turn retries/three times to retry
		0XC6	开阀堵转停止、堵转重试失败 blocked stop when opening/blocked retry failed
		0XC7	关阀堵转停止/堵转重试失败 blocked stop when closing/blocked retry failed
		0XFF	未知状态，比如上电后阀门在不确定位置， 需要执行一下开阀或关阀动作 Unknown status, such as when the valve is in an uncertain position after power is applied

入网成功后，阀门未操作，并且已经到位的情况下，阀门自动进去休眠状态。每 20 分钟唤醒一次发送一下当前状态，同时也接收服务器需要下发的数据。

After successful access to the network, the valve is not operated and has been in full opened / closed, the valve will automatically enter the dormant state. Wake up every 20 minutes to send the current status, and also receive the data that the server needs to deliver.

长按 KEY 2 秒后进行入网，红灯开始闪烁，入网成功后红灯灭。只要入网一次，后续每次上电都会自动入网。

Press KEY for 2 seconds, entering the network, LED1 red light is flashing, until to enter the network successfully, LED1 red light goes out.

Once connected to the network, it is automatically connected to the network for each subsequent power-on

短按 KEY：开阀/关阀

Short Press KEY to open/close the valve

开关阀过程中绿灯闪烁，开到位后绿灯亮，关到位后绿灯灭。

During the valve is opening or closing, the LED1 green light is flashing. When the valve is fully opened, LED1 green light keeps on. When the valve is fully closed, LED1 green light goes out.